

Linear-size deletion-saturated graphs for even cycles from line graphs of cubic hypohamiltonian graphs

github.com/graph-theory-AI

Machine-generated note

9 September 2026

Abstract

Fan, Hajebi, Hajebi and Spirkl proved that for every $t \geq 5$ there is a graph with at least one edge that is C_{2t-2} -free but in which deleting any edge creates an induced C_{2t-2} ; their construction has doubly exponential size in t , and they asked (Question 1.8) whether polynomial size is possible. We observe that the line graph of any simple cubic hypohamiltonian graph on m vertices has exactly this property for C_m , with $3m/2$ vertices. Since cubic hypohamiltonian graphs exist for every even $m \geq 26$ (Goedgebeur and Zamfirescu), this gives, for every $t \geq 14$, such a graph on exactly $3t - 3$ vertices; the finitely many smaller t are covered by the source paper itself. Hence Question 1.8 has an affirmative answer with a linear polynomial. The construction handles edge deletion only, as the question asks, not edge addition.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 4](#) records the full provenance.

1 Introduction

All graphs are finite and simple. A graph G is H -free if no induced subgraph of G is isomorphic to H . Following [2], G is H -induced-saturated if G is H -free, $G - e$ has an induced subgraph isomorphic to H for every $e \in E(G)$, and $G + e$ has an induced subgraph isomorphic to H for every $e \in E(\overline{G})$. For $t \geq 3$, C_t denotes the t -vertex cycle.

Which graphs H admit H -induced-saturated graphs is open in general; for cycles, the odd case is settled and [2] records (Observation 1.5) that the icosahedron is C_4 -induced-saturated, the Cartesian product $C_5 \square C_5$ is C_6 -induced-saturated, and the dodecahedron is C_8 -induced-saturated, with a computer-found 28-vertex C_{10} -induced-saturated graph. For longer even cycles the authors prove “half” of induced saturation:

Theorem 1.1 ([2, Theorem 1.6]). *For every integer $t \geq 5$, there is a graph G_t with $E(G_t) \neq \emptyset$ such that G_t is C_{2t-2} -free but $G_t - e$ has an induced subgraph isomorphic to C_{2t-2} for every $e \in E(G_t)$.*

Their G_t has doubly exponential size in t , and they write that they “do not even know a rationale for expecting the number of vertices of G_t to be more than linear in t ”, asking:

Question 1.2 ([2, Question 1.8]). *Does there exist a polynomial f such that for every $t \geq 3$, there is a graph G on at most $f(t)$ vertices with $E(G) \neq \emptyset$, such that G is H -free but $G - e$ has an induced subgraph isomorphic to C_{2t-2} for every $e \in E(G)$?*

(Here $H = C_{2t-2}$, as in the surrounding text of [2].) To have a name for the property in question, call a graph G C_m -deletion-saturated if $E(G) \neq \emptyset$, G is C_m -free, and $G - e$ has an induced C_m for every $e \in E(G)$. Every C_m -induced-saturated graph is C_m -deletion-saturated; the converse fails (see [2, Figure 3]).

Recall that a graph F is *hypohamiltonian* if F has no Hamiltonian cycle but $F - v$ has one for every vertex v , and that the *line graph* $L(F)$ has vertex set $E(F)$, two edges of F being adjacent in $L(F)$ when they share an endpoint.

Theorem 1.3 (Main theorem). *Let $m \geq 4$ and let F be a simple cubic hypohamiltonian graph on m vertices. Then $L(F)$ is C_m -deletion-saturated and has $3m/2$ vertices. In particular, for every $t \geq 14$ there is a C_{2t-2} -deletion-saturated graph on exactly $3t - 3$ vertices.*

Proposition 1.4 (Small cases). *For every $3 \leq t \leq 13$ there is a C_{2t-2} -deletion-saturated graph A_t .*

Corollary 1.5. *Question 1.2 has an affirmative answer with a linear polynomial: with $C = \max\{|V(A_t)| : 3 \leq t \leq 13\}$, a finite absolute constant, for every $t \geq 3$ there is a C_{2t-2} -deletion-saturated graph on at most $f(t) = 3t + C$ vertices.*

Proof. For $3 \leq t \leq 13$ use A_t from Proposition 1.4, of order at most $C \leq 3t + C$. For $t \geq 14$ use the graph of Theorem 1.3, of order $3t - 3 \leq 3t + C$. $\square$

Idea of proof. For $\ell \geq 4$, induced ℓ -cycles of a line graph $L(F)$ correspond exactly to ℓ -cycles of F (Lemma 2.1). A hypohamiltonian F on m vertices has no m -cycle, so $L(F)$ has no induced C_m . An edge e of $L(F)$ is a pair of edges ca, cb of F at a common vertex c ; if cd is the third edge at c , a Hamiltonian cycle Q of $F - d$ must pass through c using ca and cb consecutively, so in $L(F)$ the m vertices $E(Q) \cup \{cd\}$ induce a C_m with the single chord e , and deleting e leaves an induced C_m (Lemma 2.2). Cubic hypohamiltonian graphs exist for every even order $m \geq 26$ [3]; with $m = 2t - 2$ this covers all $t \geq 14$. The cases $3 \leq t \leq 13$ are supplied by [2] itself: Observation 1.5 for $t = 3, 4$ and Theorem 1.1 for $5 \leq t \leq 13$. [full proof]

Concurrent work. After the writeup was produced, the referee's literature search found that Choi [1] (arXiv, 25 August 2026) constructs a C_{2q+2} -induced-saturated graph on $8q$ vertices for every $q \geq 3$. Since induced saturation implies deletion saturation, Choi's result also answers Question 1.2 affirmatively, with $8t - 16$ vertices for $t \geq 5$. The construction here is different (line graphs of hypohamiltonian graphs, no block structure), is self-contained apart from the existence of cubic hypohamiltonian graphs, gives deletion saturation only, and is smaller: $3t - 3 < 8t - 16$ for every $t \geq 4$. See Remark 3.4.

2 The two lemmas

Lemma 2.1 (Induced cycles in line graphs). *Let F be a simple graph and $\ell \geq 4$. Then $L(F)$ has an induced cycle of length ℓ if and only if F has a cycle of length ℓ .*

Idea of proof. The edges of an ℓ -cycle of F induce a C_ℓ in $L(F)$, since consecutive edges share a vertex and non-consecutive ones are disjoint. Conversely, if $e_1, \dots, e_\ell$ induce a C_ℓ in $L(F)$, let v_i be the common vertex of e_i and e_{i+1} ; if $v_{i-1} = v_i$ then e_{i-1} and e_{i+1} would be adjacent in $L(F)$ though non-consecutive (here $\ell \geq 4$ is used), so $e_i = v_{i-1}v_i$, and the v_i are distinct for the same reason; thus $v_1 \dots v_\ell v_1$ is a cycle of F . [full proof]

Lemma 2.2 (The construction). *Let $m \geq 4$ and let F be a simple cubic hypohamiltonian graph on m vertices. Then $G = L(F)$ satisfies: G is C_m -free; $G - e$ contains an induced C_m for every $e \in E(G)$; and $|V(G)| = 3m/2$.*

Idea of proof. C_m -freeness is Lemma 2.1 plus non-Hamiltonicity. For the deletion property, write $e = xy$ with $x = ca, y = cb$ edges of F at c , and $z = cd$ the third edge at c ; take a Hamiltonian cycle Q of $F - d$. In $F - d$ the vertex c has degree exactly 2, so Q uses x and y consecutively. The set $S = E(Q) \cup \{z\}$ has m elements; $E(Q)$ induces a C_{m-1} in $L(F)$, and z is adjacent within S exactly to x and y . So $L(F)[S]$ is a C_m with the unique chord e , and $(L(F) - e)[S] \cong C_m$. The order is $|E(F)| = 3m/2$ by the handshake lemma. [full proof]

3 Remarks

Remark 3.1 (Parity). $m = 2t - 2$ is even, so $3m/2 = 3t - 3$ is an integer and a cubic graph on m vertices can exist; [Theorem 1.3](#) is stated for even m only because cubic graphs of odd order do not exist. This was implicit in the writeup.

Remark 3.2 (The additive constant). The proof of [Corollary 1.5](#) bounds C by the orders of the graphs supplied by [\[2\]](#) for $t \leq 13$; via [Theorem 1.1](#) these are doubly exponential in the fixed values $t \leq 13$, so C is finite but enormous. This is irrelevant to the truth of the corollary, and the referee notes that C can be made small: cubic hypohamiltonian graphs also exist at $m \in \{10, 18, 20, 22\}$ [\[3\]](#), covering $t \in \{6, 10, 11, 12\}$ with $3t - 3$ vertices; [Observation 1.5](#) of [\[2\]](#) covers $t \in \{3, 4, 5\}$ with at most 25 vertices and its computer-verified 28-vertex graph covers $t = 6$; only $t \in \{7, 8, 9, 13\}$ then need [Theorem 1.1](#). The writeup did not optimise this and did not need to.

Remark 3.3 (Deletion only). The construction resolves exactly what [Question 1.2](#) asks, edge deletion. In general $L(F)$ is not C_m -induced-saturated: adding a non-edge need not create an induced C_m . This limitation is stated in the writeup and is not over-claimed.

Remark 3.4 (Concurrent literature). Choi [\[1\]](#) proves that C_{2q+2} -induced-saturated graphs exist for every $q \geq 3$, on $8q$ vertices, by a construction that uses neither line graphs nor hypohamiltonian graphs and does not mention [Question 1.2](#) or the size of the graphs as a goal. Because full induced saturation implies deletion saturation, the affirmative answer to [Question 1.2](#) is now also an immediate corollary of [\[1\]](#), with a linear bound. The referee’s report records this and notes that, depending on one’s convention for near-simultaneous subsuming work, the yes/no answer could be classified as already known; the specific $3(t - 1)$ -vertex deletion-only construction appears to be new and is smaller than Choi’s for every $t \geq 4$.

Remark 3.5 (Computational checks). The referee verified [Lemma 2.1](#) exhaustively on 30 random graphs, and [Lemma 2.2](#) end to end on the line graphs of the Petersen graph ($m = 10$, 15 vertices, 30 edge deletions), the flower snark J_5 ($m = 20$, 30 vertices, 60 deletions) and the flower snark J_7 ($m = 28$, 42 vertices, 84 deletions), re-proving hypohamiltonicity by brute force and enumerating all chordless cycles; it also verified by exhaustive chordless-cycle enumeration that the icosahedron is C_4 -deletion-saturated and $C_5 \square C_5$ is C_6 -deletion-saturated. No claimed property failed ([Appendix B](#)).

Remark 3.6 (Novelty). The referee found no earlier publication containing this construction; the source paper [\[2\]](#) does not mention hypohamiltonian graphs. See [Remark 3.4](#) for the concurrent result of Choi. None of this has been checked by a human.

4 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model gpt-5.6-so1 in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id 2505.24100__01; original writeup in attacks/2505.24100__01/output.md). (2) An independent adversarial referee agent (claude-fable-5) re-derived every step, checked the cited sources, and verified the construction by exhaustive computation on the line graphs of the Petersen graph and of the flower snarks J_5 and J_7 , and the two small-case graphs of [Observation 1.5](#) of the source paper; its verdict was MINOR_GAPS, the single defect being a citation slip for $t = 3, 4$. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by claude-fable-5.1. Repairs, all taken from the referee’s report and re-checked against the source paper [\[2\]](#): (i) the small cases $t = 3, 4$ are now covered by [Observation 1.5](#) of [\[2\]](#) instead of [Theorem 1.6](#), which is stated only for $t \geq 5$ ([Proposition 1.4](#)); (ii) the parity of $m = 2t - 2$, used implicitly, is made explicit; (iii) the concurrent paper of Choi [\[1\]](#), which independently implies the yes/no answer to [Question 1.8](#), is reported ([Section 1](#), [Remark 3.4](#)); (iv) the referee’s observation that the additive constant can be made small is recorded as a remark. The text was reorganised with full proofs in [Appendix A](#). No other mathematical content was

changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

A Full proofs

Lemma 2.1 (Induced cycles in line graphs). *Let F be a simple graph and $\ell \geq 4$. Then $L(F)$ has an induced cycle of length ℓ if and only if F has a cycle of length ℓ .*

Proof. Suppose $e_1, \dots, e_\ell$ are the edges of a cycle of F , in cyclic order. Consecutive edges share an endpoint, so they are adjacent in $L(F)$; non-consecutive edges of a cycle are vertex-disjoint, so they are non-adjacent in $L(F)$. Hence $\{e_1, \dots, e_\ell\}$ induces a C_ℓ in $L(F)$.

Conversely, suppose $e_1, \dots, e_\ell$ induce a C_ℓ in $L(F)$, indices modulo ℓ , with e_i adjacent to $e_{i\pm 1}$ only. Since F is simple, two distinct adjacent edges of F share exactly one vertex; let v_i be the common vertex of e_i and e_{i+1} . We claim $v_{i-1} \neq v_i$. Otherwise e_{i-1} and e_{i+1} both contain v_i , hence are adjacent in $L(F)$; but they are distinct and non-consecutive on the cycle (this is where $\ell \geq 4$ is needed), a contradiction. Thus e_i contains the two distinct vertices v_{i-1} and v_i , i.e. $e_i = v_{i-1}v_i$. Next, the vertices $v_1, \dots, v_\ell$ are pairwise distinct: if $v_i = v_j$ with $i \neq j$, then each of e_i, e_{i+1} meets each of e_j, e_{j+1} at that vertex, and among these four edges some two are non-consecutive on the cycle (for $j = i + 1$ take e_i and e_{i+2} , distinct since $\ell \geq 4$), giving an adjacency in $L(F)$ that the induced C_ℓ forbids. Hence $v_1v_2 \dots v_\ell v_1$ is a cycle of length ℓ in F . $\square$

Lemma 2.2 (The construction). *Let $m \geq 4$ and let F be a simple cubic hypohamiltonian graph on m vertices. Then $G = L(F)$ satisfies: G is C_m -free; $G - e$ contains an induced C_m for every $e \in E(G)$; and $|V(G)| = 3m/2$.*

Proof. C_m -freeness. A cycle of length m in the m -vertex graph F would be a Hamiltonian cycle, which F does not have. By Lemma 2.1 (applicable since $m \geq 4$), $L(F)$ has no induced C_m .

Deletion property. Let e be an edge of $L(F)$. Its ends are two edges of F sharing exactly one vertex c (as F is simple); write them $x = ca$ and $y = cb$ with $a \neq b$. Since F is cubic, c has exactly one further incident edge $z = cd$, and $d \notin \{a, b, c\}$. As F is hypohamiltonian, $F - d$ has a Hamiltonian cycle Q . In $F - d$ the vertex c has degree exactly 2, its incident edges being x and y ; a Hamiltonian cycle passes through c using two of its incident edges, so Q contains both x and y , consecutively.

Put $S = E(Q) \cup \{z\} \subseteq V(L(F))$. Since Q has $m - 1$ edges and $z \notin E(Q)$ (because z is incident with $d \notin V(Q)$), $|S| = m$. By Lemma 2.1 (forward direction, length $m - 1 \geq 3$), $E(Q)$ induces a cycle C_{m-1} in $L(F)$, on which x and y are consecutive since they share c . Within S , the vertex z is adjacent exactly to x and y : it meets x and y at c ; no edge of Q is incident with d ; and x, y are the only edges of Q incident with c . Hence $L(F)[S]$ is a C_{m-1} together with a vertex z joined to the two ends x, y of the cycle edge $xy = e$; equivalently, it is an m -cycle $xzy \dots x$ with the single chord e . Therefore $(L(F) - e)[S] \cong C_m$, i.e. $L(F) - e$ contains an induced C_m .

Order. By the handshake lemma, $|V(L(F))| = |E(F)| = 3m/2$. $\square$

Theorem 1.3 (Main theorem). *Let $m \geq 4$ and let F be a simple cubic hypohamiltonian graph on m vertices. Then $L(F)$ is C_m -deletion-saturated and has $3m/2$ vertices. In particular, for every $t \geq 14$ there is a C_{2t-2} -deletion-saturated graph on exactly $3t - 3$ vertices.*

Proof. The first statement is Lemma 2.2 together with $E(L(F)) \neq \emptyset$ (a cubic graph on $m \geq 4$ vertices has adjacent edges). For the second, let $t \geq 14$ and $m = 2t - 2 \geq 26$, an even number. Goedgebeur and Zamfirescu [3, Theorem 3.6] prove that a hypohamiltonian snark of order n exists if and only if $n \in \{10, 18, 20, 22\}$ or $n \geq 26$ is even; snarks are simple cubic graphs, so there is a simple cubic hypohamiltonian graph F_m on m vertices. By the first statement, $L(F_m)$ is C_{2t-2} -deletion-saturated with $3m/2 = 3t - 3$ vertices. $\square$

Proposition 1.4 (Small cases). *For every $3 \leq t \leq 13$ there is a C_{2t-2} -deletion-saturated graph A_t .*

Proof. For $5 \leq t \leq 13$, [Theorem 1.1](#) (Theorem 1.6 of [2], valid for all $t \geq 5$) provides $A_t = G_t$. For $t = 3$ and $t = 4$, Theorem 1.6 does not apply (the original writeup cited it for these two values; this is the citation slip repaired here). Instead, Observation 1.5 of [2] states that the icosahedron is C_4 -induced-saturated and that $C_5 \square C_5$ is C_6 -induced-saturated. An H -induced-saturated graph is by definition H -free, with $G - e$ containing an induced H for every edge e , and it has edges (the icosahedron has 30, $C_5 \square C_5$ has 50); so it is H -deletion-saturated. Take A_3 the icosahedron ($2t - 2 = 4$) and $A_4 = C_5 \square C_5$ ($2t - 2 = 6$). The referee verified both deletion properties by exhaustive computation ([Appendix B](#)). $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-03) on the *original* writeup. Its step numbers refer to that writeup, not to this note; the item it labels GAP (step 8, the $t = 3, 4$ citation) is repaired in [Proposition 1.4](#). The scripts it mentions are in `verification/scripts/2505.24100__01/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The writeup formalizes the problem as: find a polynomial f such that for every $t \geq 3$ there is a graph G on at most $f(t)$ vertices with $E(G)$ nonempty, G has no induced C_{2t-2} , and $G - e$ has an induced C_{2t-2} for every $e \in E(G)$.

I fetched the source paper (arXiv:2505.24100v2, HTML full text). Question 1.8 there reads verbatim: “Does there exist a polynomial f such that for every $t \geq 3$, there is a graph G on at most $f(t)$ vertices with $E(G) \neq \emptyset$, such that G is H -free but $G - e$ has an induced subgraph isomorphic to C_{2t-2} for every $e \in E(G)$?” ($H = C_{2t-2}$ from context). This is exactly the writeup’s formalization: the deletion-only (“halfway”) property, not full induced saturation. The writeup explicitly notes at the end that it addresses only edge deletion, “exactly as in the question” — which is correct. No interpretation gaming: a reasonable author of arXiv:2505.24100 would consider Question 1.8 answered affirmatively (and optimally, up to the constant factor: linear in t , matching their remark that they “do not even know a rationale for expecting the number of vertices of G_t to be more than linear in t ”).

interpretation_ok: true.

B.2 Step-by-step findings

step	label	note
1. Statement block ($f(t) = 3t + C$; exactly $3m/2 = 3t-3$ vertices for even $m \geq 26$)	VALID	Arithmetic checks: $m = 2t-2$ gives $3m/2 = 3(t-1) = 3t-3$; m even so $3m/2$ is an integer; a cubic graph on m vertices requires m even, satisfied. Consistent with the JSON <code>one_line</code> (“ $3(t-1)$ ”).

step	label	note
2. Lemma 1, forward direction (an l -cycle in F yields an induced C_l in $L(F)$)	VALID	Consecutive edges of a cycle share an endpoint (adjacent in $L(F)$); nonconsecutive edges of a cycle are vertex-disjoint (nonadjacent in $L(F)$). Holds for all $l \geq 3$, a fortiori $l \geq 4$. Verified computationally (see below).
3. Lemma 1, converse (an induced C_l in $L(F)$, $l \geq 4$, comes from an l -cycle in F)	VALID	Re-derived in full. F simple implies each pair of adjacent line-graph vertices e_i, e_{i+1} shares a unique vertex v_i . If $v_{i-1} = v_i$ then e_{i-1} and e_{i+1} share that vertex, hence are adjacent in $L(F)$ though nonconsecutive on the induced cycle (they are distinct and nonconsecutive because $l \geq 4$) — contradiction, so $e_i = v_{i-1}v_i$. Any repetition $v_i = v_j$ forces two nonconsecutive edges among $e_i, e_{i+1}, e_j, e_{j+1}$ to share a vertex (for $j = i+1$ the pair e_i, e_{i+2} works since $l \geq 4$), again a contradiction. So $v_1 \dots v_l v_1$ is an l -cycle. This is the classical fact that induced cycles of length ≥ 4 in a line graph correspond to cycles of the root graph (only triangles have a second origin, $K_{1,3}$); the $l \geq 4$ hypothesis is used and is available since $m = 2t-2 \geq 4$ for $t \geq 3$. Also verified exhaustively on 30 random graphs by machine.
4. Lemma 2, C_m -freeness of $L(F)$	VALID	An m -cycle in an m -vertex graph is a Hamiltonian cycle; F hypohamiltonian is non-Hamiltonian, so F has no C_m , so by Lemma 1 ($m \geq 4$) $L(F)$ has no induced C_m . Verified by exhaustive chordless-cycle enumeration on three concrete graphs (see Computational check).

step	label	note
5. Lemma 2, deletion property (witness $S = E(Q) \cup \{z\}$)	VALID	Re-derived in full. Every edge e of $L(F)$ is a pair of F -edges $x = ca$, $y = cb$ sharing exactly one vertex c (F simple); cubicity gives a unique third edge $z = cd$ at c , with d not in $\{a, b, c\}$. $F - d$ has a Hamiltonian cycle Q (hypohamiltonicity); c has degree exactly 2 in $F - d$ with incident edges x, y , so Q traverses x, y consecutively, making $xy = e$ an edge of the C_{m-1} that $E(Q)$ induces in $L(F)$ (Lemma 1 forward, length $m-1 \geq 3$). z is not in $E(Q)$ (z is incident with d), so =
6. Lemma 2, order count	$V(L(F))$	
7. Section 3, existence of cubic hypohamiltonian graphs on every even $m \geq 26$	VALID	The cited result exists and states exactly what is used: Goedgebeur–Zamfirescu, “On hypohamiltonian snarks and a theorem of Fiorini” (arXiv:1608.07164; Ars Math. Contemp.), Theorem 3.6: “There exists a hypohamiltonian snark of order n iff n in $\{10, 18, 20, 22\}$ or n is even and $n \geq 26$.” Snarks are by definition simple cubic graphs, so the writeup’s quoted spectrum $\{10, 18, 20, 22\} \cup \{\text{even } n \geq 26\}$ and the “eventual existence” part it needs are both correct verbatim. For $t \geq 14$, $m = 2t-2 \geq 26$, so F_m exists and

step	label	note
8. Section 4, small cases $3 \leq t \leq 13$ via “Theorem 1.6 of the source paper”	GAP (repaired)	Citation slip: the source paper states Theorem 1.6 only for $t \geq 5$ (“For every integer $t \geq 5$, there is a graph G_t with $E(G_t) \neq \text{emptyset}$ such that G_t is C_{2t-2} -free but $G_t - e$ has an induced subgraph isomorphic to C_{2t-2} for every $e \in E(G_t)$ ”). So Theorem 1.6 does NOT supply A_3 (C_4) or A_4 (C_6) as the writeup asserts. The repair is routine and stays inside the same paper: Observation 1.5 states the icosahedron is C_4 -induced-saturated and $C_5 \times C_5$ is C_6 -induced-saturated; induced saturation implies the deletion-only property. I verified both graphs’ deletion property by brute force (see Computational check). For $5 \leq t \leq 13$ Theorem 1.6 applies as cited.
9. Conclusion, $f(t) = 3t + C$ linear, $C = \max$ of the finitely many small-case orders	VALID	For $3 \leq t \leq 13$,

B.3 Reference check

1. **Goedgebeur & Zamfirescu, “On hypohamiltonian snarks and a theorem of Fiorini”** (arXiv:1608.07164, Ars Mathematica Contemporanea). Fetched the full PDF. Theorem 3.6 states verbatim: “There exists a hypohamiltonian snark of order n iff n in $\{10, 18, 20, 22\}$ or n is even and $n \geq 26$.” Table 1 of that paper reports 95 hypohamiltonian snarks of order 26. Snark = cubic, cyclically 4-edge-connected, chromatic index 4, girth ≥ 5 — in particular a simple cubic hypohamiltonian graph. Exactly what the writeup uses (it only needs the “even $n \geq 26$ ” direction). **Checks out.**
2. **Theorem 1.6 of the source paper** (arXiv:2505.24100v2, fetched HTML full text). Exists and gives the deletion-only property for C_{2t-2} — but for $t \geq 5$, not $t \geq 3$ as the writeup’s Section 4 uses. **Mis-cited at $t = 3, 4$** (the reason for references_ok: false). Repairable via the same paper’s Observation 1.5: “The icosahedron is C_4 -induced-saturated, the Cartesian product of two 5-cycles is C_6 -induced-saturated, and the dodecahedron is C_8 -induced-saturated” (plus a computer-verified C_{10} example), which is stronger than what is needed.
3. **Novelty / prior-art search.** The source paper itself nowhere mentions hypohamiltonian graphs (checked in the full text); its own construction is doubly exponential. A web search for the combination induced-saturated / even cycle / line graph / hypohamiltonian found no paper containing this construction. However, it found **Ilkyoo Choi, “Induced-saturated graphs exist for even cycles”** (arXiv:2608.24202, submitted 2026-08-25), which constructs, for every $q \geq 3$, a C_{2q+2} -induced-saturated graph on $8q$ vertices by a block construction (no

line graphs, no hypohamiltonian graphs, and no mention of Question 1.8 or of size bounds as a goal). Since full induced saturation implies the deletion-only property, Choi's result makes the affirmative answer to Question 1.8 an immediate corollary of the very recent literature (linear: $8q = 8t - 16$ vertices for $t \geq 5$). The writeup's construction remains different, self-contained, deletion-only, and smaller ($3t - 3 < 8t - 16$ for all $t \geq 4$).

B.4 Computational check

Scripts (reproducible):

- `verification/scripts/2505.24100__01/verify.py`
- `verification/scripts/2505.24100__01/verify_small_t.py`

Run with `uv run --with networkx python3 <script>` (networkx 3.6.1, Python 3.14.6). Total runtime under 3 seconds. All checks passed. Exact numbers:

1. **Lemma 1** verified exhaustively on 30 random simple graphs (7-11 vertices, edge probability 0.2-0.45): the set of lengths $l \geq 4$ of cycles of F equals the set of lengths $l \geq 4$ of chordless cycles of $L(F)$, in every trial.
2. **Lemma 2** verified end-to-end on three concrete simple cubic graphs, with hypohamiltonicity itself re-proved by brute-force Hamiltonian-cycle search (F non-Hamiltonian, $F - v$ Hamiltonian for every v), C_m -freeness by exhaustive chordless-cycle enumeration (`networkx chordless_cycles` with `length_bound = m`), and the deletion property for every single edge of $L(F)$ by directly checking that the witness set induces a connected 2-regular graph on m vertices in $L(F) - e$:
 - **Petersen graph** ($m = 10, t = 6$): $L(F)$ has $15 = 3m/2$ vertices; 67 chordless cycles of length ≤ 10 , none of length 10; all 30 edge deletions create an induced C_{10} — confirmed both by the witness and by an independent exhaustive chordless-cycle search in each of the 30 graphs $L(F) - e$.
 - **Flower snark J_5** ($m = 20, t = 11$): $30 = 3m/2$ vertices; 1464 chordless cycles of length ≤ 20 , none of length 20; all 60 edge deletions create an induced C_{20} .
 - **Flower snark J_7** ($m = 28, t = 15$, inside the writeup's $t \geq 14$ regime): $42 = 3m/2$ vertices; 18848 chordless cycles of length ≤ 28 , none of length 28; all 84 edge deletions create an induced C_{28} .
3. **Repair of step 8** ($t = 3, 4$): the icosahedron (12 vertices) has no induced C_4 and every one of its 30 edge deletions creates an induced C_4 ; $C_5 \times C_5$ (25 vertices) has no induced C_6 and every one of its 50 edge deletions creates an induced C_6 . Both by exhaustive chordless-cycle enumeration.

No claimed property failed.

B.5 Caveats

1. **The $t = 3, 4$ citation slip** (step 8): “Theorem 1.6 ... for $3 \leq t \leq 13$ ” is false as written for $t = 3, 4$ (Theorem 1.6 requires $t \geq 5$). Fully repaired above via Observation 1.5 of the same source paper and verified by machine; this is the only reason the verdict is MINOR_GAPS rather than CONFIRMED.
2. **Concurrent literature.** Choi (arXiv:2608.24202, 2026-08-25, nine days before this review) proves the strictly stronger full induced-saturation statement for every even cycle $C_{\{2q+2\}}$, $q \geq 3$, on $8q$ vertices. The affirmative answer to Question 1.8 is therefore now derivable from the literature,

though Choi does not state Question 1.8, does not discuss size as a goal, and uses an entirely different construction. Depending on the ledger’s convention for near-simultaneous subsuming work, ALREADY_KNOWN could be argued for the yes/no answer; the specific $3(t-1)$ -vertex deletion-only construction via line graphs of hypohamiltonian cubic graphs appears to be new and is smaller than Choi’s for every $t \geq 4$.

3. **Deletion only.** The construction resolves exactly the deletion half asked by Question 1.8; $L(F)$ is generally NOT C_m -induced-saturated (adding a nonedge need not create an induced C_m). The writeup states this limitation explicitly, so nothing is over-claimed.
4. **Parity.** The argument silently uses that $m = 2t-2$ is even (a cubic graph on an odd number of vertices does not exist, and $3m/2$ must be an integer); this is automatic here but worth noting as a convention dependence.
5. **The constant C is astronomically large** if instantiated via Theorem 1.6 for $7 \leq t \leq 13$ (those A_t are doubly exponential in the fixed t), so $f(t) = 3t + C$ is linear with an enormous additive constant. This is irrelevant to the truth of the claim (any fixed C is a constant), and in fact C can be made small: cubic hypohamiltonian graphs also exist at $m = 10, 18, 20, 22$ (covering $t = 6, 10, 11, 12$), and Observation 1.5 covers $t = 3, 4, 5, 6$ with graphs of at most 25 vertices, leaving only t in $\{7, 8, 9, 13\}$ to Theorem 1.6. The writeup did not optimize this; it did not need to.
6. **Boundary case $m = 26$ ($t = 14$)** was verified through the literature (Goedgebeur–Zamfirescu Theorem 3.6 and their Table 1 listing 95 hypohamiltonian snarks of order 26), not by machine; my machine checks instantiate the construction at $m = 10, 20, 28$. The proof of Lemma 2 is uniform in m , so this affects nothing.

B.6 Referee summary

I set out to break this proof and could not. The construction — take the line graph of any simple cubic hypohamiltonian graph on $m = 2t-2$ vertices — is short, correct, and verifiably answers Question 1.8 of arXiv:2505.24100 affirmatively with a linear polynomial, giving graphs on exactly $3t-3$ vertices for every $t \geq 14$ (and the general statement for all $t \geq 3$ after absorbing finitely many small cases into a constant). Every step re-derives cleanly: the classical correspondence between induced cycles of length ≥ 4 in a line graph and cycles of the root graph, the degree-2 forcing of the two root edges through c in $F - d$, and the single-chord structure of the witness set. Every claimed property was independently confirmed by brute-force computation on three concrete cubic hypohamiltonian graphs (Petersen, J_5, J_7), including exhaustive chordless-cycle enumeration and per-edge deletion checks, and the one external citation (Goedgebeur–Zamfirescu’s spectrum theorem for hypohamiltonian snarks) states verbatim what is used. The single defect is a citation slip: Theorem 1.6 of the source paper is stated for $t \geq 5$, not $t \geq 3$, so the cases $t = 3, 4$ need Observation 1.5 of the same paper instead — a one-line repair, machine-verified here. Separately, note that Choi (arXiv:2608.24202, Aug 25, 2026) has just published a stronger full induced-saturation construction ($8q$ vertices) that independently implies the affirmative answer, though without addressing Question 1.8 or size; the writeup’s construction is different and smaller.

References

- [1] I. Choi, Induced-saturated graphs exist for even cycles, arXiv:2608.24202 (25 August 2026).
- [2] X. Fan, S. Hajebi, S. Hajebi and S. Spirkl, Halfway to induced saturation for even cycles, arXiv:2505.24100 (2025; v2 June 2025).
- [3] J. Goedgebeur and C. T. Zamfirescu, On hypohamiltonian snarks and a theorem of Fiorini, *Ars Mathematica Contemporanea*; arXiv:1608.07164.